AS · Unit 2 · WPH12 Waves & Electricity

Superposition & Stationary Waves

What happens when two waves meet? They simply add up, point by point. That one idea, superposition, explains the loud and quiet spots between two loudspeakers, the bright and dark bands in a ripple tank, and the stationary waves on a guitar string that decide which notes it can play. We'll build it all up from the idea of phase.

Phase and phase difference

Every particle on a progressive wave oscillates with the same frequency, but they don't all reach the top at the same moment. A point a little further along the wave does exactly what the point behind it did, just slightly later. **Phase** is the word for "where a point is in its cycle", and we measure it as an angle: one complete cycle is 2π radians, or 360°.

DEFINITION

Phase, phase difference and wavefronts

- **Phase:** the position of a point within its cycle of oscillation, given as an angle (one cycle = 2π rad = 360°).
- **Phase difference:** the fraction of a cycle by which one oscillation is ahead of or behind another, expressed as an angle in radians or degrees.
- **In phase:** phase difference of zero (or a whole number of cycles: $2\pi, 4\pi, \ldots$). **Antiphase:** phase difference of π rad (180°): one point is at a crest when the other is at a trough.
- **Wavefront:** a line or surface joining points on a wave that are in phase, for example all the points along one crest. Wavefronts are usually drawn one wavelength apart (crest to crest), and the wave travels at right angles to them.

Along a progressive wave the phase changes steadily with distance. Walk one whole wavelength along the wave and you have gone through one complete cycle, so you are back in phase. That means the phase difference between two points depends only on **what fraction of a wavelength apart** they are. Exactly the same rule applies to two waves arriving at one point after travelling different distances: the extra distance one of them travels (the **path difference**) shifts its phase in the same way.

Phase difference from path difference

$$\phi = \frac{2\pi\,\Delta x}{\lambda} \qquad \text{or} \qquad \phi = 360^\circ \times \frac{\Delta x}{\lambda}$$

ϕ: phase difference / rad (or °) Δx: separation along the wave, or path difference / m
λ: wavelength / m

Relating phase difference to path difference

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 One wavelength along a progressive wave corresponds to one complete cycle of phase:

$$\Delta x = \lambda \quad \longleftrightarrow \quad \phi = 2\pi \text{ rad}$$

2 A distance Δx is the fraction $\Delta x/\lambda$ of a wavelength, so it gives the same fraction of a full cycle:

$$\frac{\phi}{2\pi} = \frac{\Delta x}{\lambda}$$

3 Multiply both sides by 2π:

$$\boxed{\phi = \frac{2\pi\,\Delta x}{\lambda}}$$

Check the special cases: $\Delta x = \lambda/2$ gives $\phi = \pi$ (antiphase); $\Delta x = \lambda$ gives 2π (back in phase).

EXAM TIP

Give phase differences in radians as multiples of π: $\pi/2$, π, $3\pi/2$. If two points are more than one wavelength apart, take off whole wavelengths first: a separation of 1.25λ gives the same phase difference as 0.25λ, which is $\pi/2$. Mark schemes accept "antiphase", "π rad" or "180°".

WORKED EXAMPLE

A progressive wave has frequency 25 Hz and speed 15 m s^{-1}. Points P and Q lie 0.45 m apart along the direction the wave travels.

(a) Calculate the phase difference between P and Q.
(b) State the smallest distance from P to a point that oscillates in antiphase with P.

Show solution

(a) First find the wavelength from $v = f\lambda$:

$$\lambda = \frac{v}{f} = \frac{15}{25} = 0.60 \text{ m}$$

Then the phase difference:

$$\phi = \frac{2\pi\,\Delta x}{\lambda} = \frac{2\pi \times 0.45}{0.60} = 1.5\pi \text{ rad} = \frac{3\pi}{2} \text{ rad } (270^\circ)$$

Q is further along, so Q lags P by $3\pi/2$ rad. (You could equally say Q leads P by $\pi/2$ rad: lagging by three quarters of a cycle looks the same as leading by one quarter.)

(b) Antiphase means $\phi = \pi$, which is half a wavelength: $\lambda/2 = 0.30$ m.

The principle of superposition

Drop two stones in a pond and the ripples pass straight through each other, carrying on afterwards as if nothing had happened. While they overlap, though, the water surface has to do something. The rule is beautifully simple: each wave tries to displace the water, and the actual displacement is the sum of the two.

DEFINITION

Principle of superposition

When two or more waves meet at a point, the resultant displacement at that point is the (vector) sum of the displacements that each wave would cause on its own.

- **Constructive superposition:** the waves arrive **in phase**, crest on crest. The amplitudes add: $A = A_1 + A_2$.
- **Destructive superposition:** the waves arrive in **antiphase**, crest on trough. The amplitudes subtract: $A = |A_1 - A_2|$. If the amplitudes are equal, the waves cancel completely.
- Any other phase difference gives something in between.

COMMON MISTAKE

Displacements add **with their signs**. A displacement of +1.2 cm from one wave and −0.8 cm from the other gives +0.4 cm, not 2.0 cm. And destructive interference doesn't destroy energy: where one region is quiet,

the energy turns up in the loud regions nearby.

Coherence, path difference and interference

Superposition happens whenever waves meet, but usually you don't notice: the pattern of loud and quiet spots shifts around so quickly that it averages out. To see a **steady** pattern, the phase difference between the two sources must stay the same. That is what coherence means.

DEFINITION

Coherence, path difference and interference

- **Coherent sources:** sources that have the same frequency and a constant phase difference.
- **Path difference:** the difference between the distances travelled by two waves from their sources to a given point.
- **Interference:** the superposition of waves from coherent sources, which produces a stable pattern of maxima (constructive) and minima (destructive).

The easiest way to get coherent sources is to make one source do the job of two: two loudspeakers driven by the **same signal generator**, two dippers on the **same vibrating bar** in a ripple tank, or laser light passing through two slits. Two separate light bulbs are never coherent. Each atom emits light in short, random bursts, so the phase difference between the bulbs changes randomly and extremely rapidly, and any pattern washes out.

Now think about a point P in front of two coherent sources that are in phase. Whatever happens at P depends on how much further one wave has travelled than the other. Using $\phi = 2\pi\Delta x/\lambda$:

KEY IDEA

Conditions for maxima and minima (sources in phase)

Path difference	Phase difference at P	Result
$0,\ \lambda,\ 2\lambda,\ \cdots = n\lambda$	$0,\ 2\pi,\ 4\pi$ … (in phase)	constructive: **maximum**
$\frac{1}{2}\lambda,\ \frac{3}{2}\lambda,\ \cdots = (n+\frac{1}{2})\lambda$	$\pi,\ 3\pi$ … (antiphase)	destructive: **minimum**

If the two sources are themselves in antiphase, everything swaps over: a path difference of $n\lambda$ gives a minimum and $(n+\frac{1}{2})\lambda$ gives a maximum.

For a minimum to be completely silent (or completely dark), the two waves must also have **equal amplitudes** when they arrive, and transverse waves such as light must be polarised in the same plane. In real experiments the

minima are usually just quieter, not silent, because the waves have spread out by different amounts and walls reflect extra sound in.

WORKED EXAMPLE

Two loudspeakers A and B are connected to the same signal generator and emit sound in phase at 850 Hz. A microphone is 2.60 m from A and 3.20 m from B. The speed of sound is 340 m s^{-1}.
(a) Show that the microphone is at a minimum.
(b) The frequency is increased slowly. Find the next frequency at which the microphone detects a maximum.
(c) Suggest why the sound at the minimum does not drop to zero.

Show solution

(a) Wavelength: $\lambda = v/f = 340/850 = 0.400\ \text{m}$. Path difference: $3.20 - 2.60 = 0.60\ \text{m}$. In wavelengths:

$$\frac{\Delta x}{\lambda} = \frac{0.60}{0.400} = 1.5$$

A path difference of 1.5λ means the waves arrive in antiphase ($\phi = 3\pi$), so there is destructive interference: a minimum.

(b) A maximum needs $\Delta x = n\lambda$, so $\lambda = 0.60/n$. Increasing f makes λ smaller than 0.400 m. The next allowed value is $n = 2$:

$$\lambda = \frac{0.60}{2} = 0.30\ \text{m} \quad \Rightarrow \quad f = \frac{v}{\lambda} = \frac{340}{0.30} = 1130\ \text{Hz}\ (3\ \text{s.f.})$$

(c) The microphone is further from B, so the wave from B has spread out more and has a smaller amplitude than the wave from A. Unequal amplitudes can't cancel completely. Sound reflected off the walls and floor also reaches the microphone.

COMMON MISTAKE

A point the same distance from both sources (path difference zero) is a maximum **only if the sources are in phase**. If they are in antiphase, the waves arrive crest on trough and the centre line is a minimum.

Stationary waves: how they form

Flick one end of a rope tied to a wall and the pulse comes back, reflected. Keep shaking steadily and you now have two waves on the rope at once: the one you are sending and its reflection, with the same frequency, speed and (nearly) the same amplitude, travelling in **opposite directions**. Their superposition is something new. The

wave pattern no longer moves along the rope. Some points never move at all, while others in between swing up and down with a large amplitude. This is a **stationary** (or standing) wave.

DEFINITION

Stationary waves, nodes and antinodes

- **Stationary wave:** the wave formed when two progressive waves of the same frequency (and similar amplitude) travelling in opposite directions superpose. It stores energy rather than transferring it.
- **Node:** a point on a stationary wave with zero amplitude (no displacement at any time).
- **Antinode:** a point on a stationary wave with maximum amplitude.

KEY IDEA

Why there are nodes, and why they are λ/2 apart

At a **node**, the two travelling waves always arrive in **antiphase**, so they cancel at every instant. At an **antinode** they always arrive **in phase** and add to give twice the amplitude of either wave.

Move a distance Δx along. The wave travelling one way now has to travel Δx further and the other Δx less, so their phase difference changes by $2 \times 2\pi\Delta x/\lambda$. To get from one antiphase point to the next, that change must be 2π, which happens when $\Delta x = \lambda/2$. So **adjacent nodes are $\lambda/2$ apart**, with an antinode halfway between, $\lambda/4$ from each node.

You can make stationary waves with any kind of wave, as long as two waves of the same frequency travel through each other in opposite directions. The usual trick is **reflection**:

- a string fixed at both ends and vibrated (a guitar, or the vibration generator in Core Practical 5);
- microwaves from a transmitter reflected back by a metal plate, detected with a probe moved along the line between them;
- sound from a loudspeaker reflected from a wall or from the closed end of a tube.

Graphs of stationary waves

A stationary wave is usually drawn as an **envelope**: the two extreme positions of the string, one solid and one dashed. If a question asks for the shape at particular instants, draw snapshots. Start the clock ($t = 0$) when the string is at its full displacement. A quarter of a period later ($t = T/4$) it is momentarily **completely straight**, with every point rushing through equilibrium at its maximum speed, and at $t = T/2$ it is fully displaced the other way. The energy swaps between kinetic energy (string straight) and elastic potential energy (string at maximum displacement).

The phase pattern is completely different from a progressive wave. All the points between two neighbouring nodes reach their maximum displacement **at the same moment**, so they are in phase, even though their amplitudes are

different. Points in adjacent loops are in **antiphase**: when one loop is up, the next is down. Look at the red and blue particles in the simulation.

	Progressive wave	Stationary wave
Energy	transferred along the wave	stored, not transferred
Amplitude	the same for every point (if no energy is lost)	varies with position: zero at nodes, maximum at antinodes
Phase	changes continuously along the wave: 2π every wavelength	all points between adjacent nodes in phase; adjacent loops in antiphase
Wavelength	distance between adjacent points in phase	twice the distance between adjacent nodes
Frequency	every point oscillates at the wave's frequency	every point (except nodes) oscillates at the same frequency

EXAM TIP

Node to node = $\lambda/2$. Node to the next antinode = $\lambda/4$. Count **loops**, not nodes: n loops make n half-wavelengths. For a longitudinal stationary wave (sound in a tube) the graph looks just the same, but the displacement it shows is along the tube, not sideways.

WORKED EXAMPLE

A microwave transmitter faces a flat metal plate. A detector moved along the line between them finds a series of maxima and minima. Moving the detector from one minimum to the sixth minimum after it covers a distance of 8.4 cm.
(a) Explain why the minima occur.
(b) Calculate the frequency of the microwaves.
(c) Explain why the signal at each minimum is small but not zero.

Show solution

(a) Microwaves reflect from the plate and travel back towards the transmitter. The incident and reflected waves have the same frequency and travel in opposite directions, so they superpose to form a stationary wave. The minima are the nodes, where the two waves are always in antiphase and cancel.

(b) The starting minimum and the sixth one after it are 6 node-to-node gaps apart, and each gap is $\lambda/2$:

$$6 \times \frac{\lambda}{2} = 8.4 \text{ cm} \quad \Rightarrow \quad \lambda = 2.8 \text{ cm} = 0.028 \text{ m}$$

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8}{0.028} = 1.1 \times 10^{10}\ \text{Hz}\ (2\ \text{s.f.})$$

(c) The reflected wave has a smaller amplitude than the incident wave (some energy is absorbed and it spreads out), so the two cannot cancel completely at the nodes.

Waves on a string: v = √(T/μ)

Pluck a guitar string and a wave travels along it, reflects off the ends and sets up a stationary wave. How fast does the wave travel? It depends only on two properties of the string.

Speed of a transverse wave on a string

FORMULA SHEET

$$v = \sqrt{\frac{T}{\mu}}$$

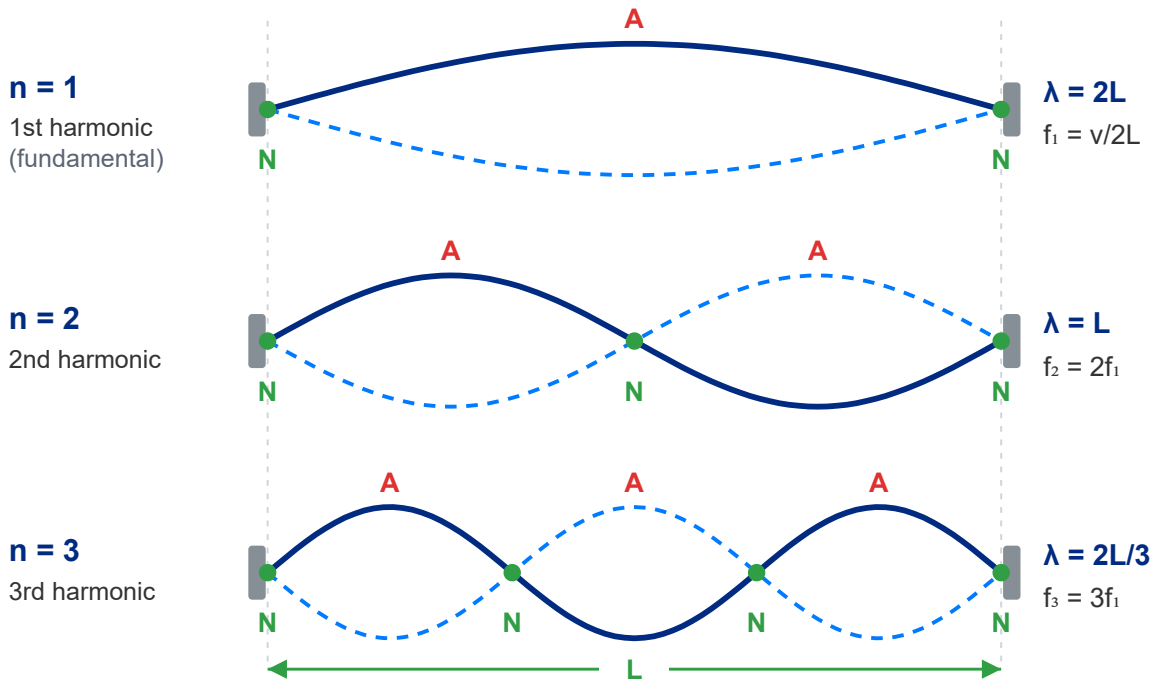
v: wave speed / m s⁻¹ T: tension / N μ: mass per unit length / kg m⁻¹

- **Tension** provides the restoring force. A tighter string snaps back more quickly when it is displaced, so the disturbance is passed on faster: v goes up with T.
- **Mass per unit length** μ is the string's inertia. A heavier string responds more sluggishly to the same force, so v goes down as μ goes up. For a wire, $\mu = m/l$ (mass ÷ length), which is also density × cross-sectional area.
- The speed does **not** depend on the frequency or amplitude of the wave. Change f and the wavelength changes instead, so that $v = f\lambda$ still holds.

A quick units check shows the equation is homogeneous: $\sqrt{\text{N/kg m}^{-1}} = \sqrt{\text{kg m s}^{-2} \times \text{m kg}^{-1}} = \sqrt{\text{m}^2\,\text{s}^{-2}} = \text{m s}^{-1}$.

Harmonics: which stationary waves fit?

A string clamped at both ends must have a **node at each end**. So a stationary wave can only form if a whole number of loops (half-wavelengths) fits exactly into the length L. Each of these patterns is a **harmonic**, and the string resonates only at their frequencies.



The first three harmonics on a string of length L. N = node, A = antinode. Each loop is half a wavelength, so n loops give $\lambda = 2L/n$.

Frequencies of the harmonics on a string

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 Both ends are nodes, and each loop is half a wavelength. With n loops on a string of length L:

$$L = n\frac{\lambda}{2} \quad \Rightarrow \quad \lambda = \frac{2L}{n}$$

2 The wave speed on the string is fixed by T and μ, so use $v = f\lambda$ to find the frequency:

$$f = \frac{v}{\lambda} = \frac{nv}{2L}$$

3 Substitute $v = \sqrt{T/\mu}$:

$$\boxed{f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}}$$

4 For the fundamental, $n = 1$:

$$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

So $f_1 \propto 1/L$, $f_1 \propto \sqrt{T}$ and $f_1 \propto 1/\sqrt{\mu}$: exactly the three relationships Core Practical 5 tests. Short, tight, thin strings give high notes.

COMMON MISTAKE

The λ in these equations is the wavelength of the wave **on the string**, not of the sound it makes in the air. The frequency is the same in both, but the sound travels at about 340 m s^{-1}, so its wavelength is different.

WORKED EXAMPLE

A guitar string has a vibrating length of 0.648 m and a mass per unit length of 1.15 × 10^{-3} kg m^{-1}. It plays a fundamental frequency of 110 Hz.
(a) Calculate the speed of waves on the string.
(b) Calculate the tension in the string.
(c) The player presses the string against a fret so that the vibrating length becomes 0.432 m. Calculate the new fundamental frequency.
(d) The tension is then increased by 10%. By what factor does the frequency change?

Show solution

(a) For the fundamental, $\lambda = 2L = 2 \times 0.648 = 1.296$ m:

$$v = f\lambda = 110 \times 1.296 = 142.6 \text{ m s}^{-1} \approx 143 \text{ m s}^{-1}$$

(b) Square $v = \sqrt{T/\mu}$ and rearrange:

$$T = \mu v^2 = 1.15 \times 10^{-3} \times 142.6^2 = 23.4 \text{ N (3 s.f.)}$$

(c) T and μ are unchanged, so v is unchanged and $f_1 \propto 1/L$:

$$f_1 = \frac{v}{2L} = \frac{142.6}{2 \times 0.432} = 165 \text{ Hz (3 s.f.)}$$

(d) $f \propto \sqrt{T}$, so the frequency changes by a factor of $\sqrt{1.10} = 1.049$: it rises by about 4.9%.

Core Practical 5: frequency of a vibrating string

CORE PRACTICAL

Investigate the effects of length, tension and mass per unit length on the frequency of a vibrating string or wire

Apparatus: a string or wire fixed at one end to a vibration generator (driven by a signal generator), passing over a bridge and a pulley at the other end, with a mass hanger hanging from it to set the tension $T = mg$. A second movable bridge can set the vibrating length L.

1. Set the length, tension and wire. Slowly increase the signal generator frequency from a low value until the string vibrates in one loop with the largest amplitude: this is the **fundamental** f_1. Read f from the signal generator.
2. **Length:** keep T and μ fixed and repeat for at least 6 lengths, measured between the bridges (nodes) with a metre rule. Plot f_1 **against** $1/L$: a straight line through the origin with **gradient** $= \frac{1}{2}\sqrt{T/\mu}$ $= v/2$.
3. **Tension:** keep L and μ fixed and add masses. Plot f_1 **against** $\sqrt{T}$: gradient $= 1/(2L\sqrt{\mu})$.
4. **Mass per unit length:** keep L and T fixed and use different wires. Find each μ by weighing a measured length of the same wire on a top-pan balance ($\mu = m/l$), or from its density and diameter measured with a micrometer. Plot f_1 **against** $1/\sqrt{\mu}$: gradient $= \sqrt{T}/(2L)$.

- **Judging resonance** is the main source of uncertainty, because the amplitude is large over a range of frequencies. Approach the resonance from below and from above, note both frequencies and average them. A strobe lamp makes it easier to see when the loops are largest. You can also find a higher harmonic and divide its frequency by n, which can reduce the percentage uncertainty in f_1.
- The end at the vibration generator is not a perfect node, so measure L between the nodes you can see, not to the generator.
- Use the mass of the hanger in T, and a low-friction pulley so the tension really is mg. The wire stretches slightly as the tension increases, which changes μ a little.
- **Safety:** wear eye protection in case the wire snaps, keep feet clear of the hanging masses (place a box of sand or a cushion underneath), and don't run the vibration generator above its rated current.

WORKED EXAMPLE

In Core Practical 5 a student keeps the tension at 4.9 N and plots f_1 against $1/L$. The best-fit line passes through the origin with gradient 47.5 Hz m. Calculate the mass per unit length of the wire.

Show solution

Since $f_1 = \dfrac{v}{2} \times \dfrac{1}{L}$, the gradient is $v/2$:

$$v = 2 \times 47.5 = 95.0 \text{ m s}^{-1}$$

Then from $v = \sqrt{T/\mu}$:

$$\mu = \frac{T}{v^2} = \frac{4.9}{95.0^2} = 5.4 \times 10^{-4} \text{ kg m}^{-1} \text{ (2 s.f.)}$$

Quick checks

QUICK CHECK

Two points on a progressive wave are 0.15 m apart along the direction of travel. The wavelength is 0.60 m. What is the phase difference between them?

- 0.25 rad
- $\pi/4$ rad
- $\pi/2$ rad
- π rad

The points are $0.15/0.60 = 0.25$ of a wavelength apart, so $\phi = 0.25 \times 2\pi = \pi/2$ rad. "0.25 rad" forgets to multiply the fraction of a cycle by 2π.

QUICK CHECK

Which of these is needed for two sources to be coherent?

- The same amplitude and the same frequency
- The same frequency and a constant phase difference
- The same amplitude and zero phase difference
- The same wavelength and the same amplitude

Coherence is about a steady phase relationship, which needs equal frequencies. The amplitudes can differ (you just won't get complete cancellation), and the phase difference can be any value as long as it doesn't change.

QUICK CHECK

Two loudspeakers are connected so that they vibrate in antiphase. A microphone is placed at a point exactly the same distance from both. What does it detect?

- A maximum, because the path difference is zero
- A minimum, because the waves arrive in antiphase
- Alternating maxima and minima
- No interference, because the sources are not coherent

With zero path difference the waves arrive with the same phase difference they left with: π. Antiphase sources are still coherent, because the phase difference is constant.

QUICK CHECK

Points X and Y lie between the same pair of adjacent nodes on a stationary wave. X is closer to the antinode. Which statement is correct?

- X and Y have the same amplitude but are out of phase.
- X and Y are in antiphase.
- X and Y are in phase but X has a larger amplitude.
- X and Y have the same amplitude and are in phase.

Within one loop of a stationary wave every point moves up and down together (in phase), but the amplitude increases from zero at the nodes to a maximum at the antinode.

QUICK CHECK

On a stationary wave of wavelength λ, what is the distance between a node and the nearest antinode?

- λ/4
- λ/2
- λ
- 2λ

Adjacent nodes are $\lambda/2$ apart and the antinode lies halfway between them, so it is $\lambda/4$ from each node.

QUICK CHECK

The tension in a vibrating string is made four times bigger. The length and the string are unchanged. What happens to the fundamental frequency?

- It is quartered.
- It doubles.
- It is four times bigger.
- It stays the same, because the length is unchanged.

The wavelength of the fundamental is fixed at $2L$, but $v = \sqrt{T/\mu}$ doubles when T is multiplied by 4. Since $f = v/\lambda$, the frequency doubles too.